

Task 2: Visual Odometry / SLAM

GitHub: <https://github.com/Pekeps/slam-project>

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Abstract—We present a stereo visual SLAM system for autonomous vehicle localization. The system uses ORB features with stereo triangulation for metric depth recovery, estimates camera motion via PnP-RANSAC, and corrects accumulated drift through loop closure detection and pose-graph optimization. Evaluated on the KITTI Odometry Benchmark (sequences 00–10), the system achieves an average translational error of 7.61 and rotational error of 0.031 deg/m across the sequences.

I. INTRODUCTION

This project implements a stereo visual SLAM system designed for localizing autonomous vehicles using visual odometry. Using a stereo camera setup, the system performs direct depth estimation through triangulation, eliminating the scaling issues found in monocular configurations.

The system estimates camera pose using the Perspective-n-Point (PnP) algorithm with RANSAC for outlier rejection. To correct odometry drift over long trajectories, the system includes loop closure detection and pose-graph optimization. When the system recognizes a previously visited location, it adds a constraint to the pose graph and re-optimizes the full trajectory, reducing accumulated positional error.

II. METHODOLOGY

The system architecture is divided into a real-time tracking front-end and a graph-based optimization back-end.

A. Feature Extraction and Stereo Matching

We use ORB features for both frame-to-frame tracking and loop closure matching. Left and right stereo images are matched using brute-force Hamming-distance matching with a ratio test, and matched points are triangulated to obtain 3D coordinates in the camera frame. This stereo triangulation provides metric depth directly, avoiding the scale drift that affects monocular systems.

B. Motion Estimation

Camera pose is estimated by solving the PnP problem: given 3D points from a reference keyframe and their 2D projections in the current frame, `cv2.solvePnP` recovers the relative pose. We use a reprojection error threshold of 3.0 pixels and a RANSAC confidence of 0.999.

C. Pose Graph

The pose graph is built from keyframes. Every 10th frame is considered as a keyframe candidate and inserted into the graph as a vertex if it contains valid features.

D. Loop Closure Detection

The system checks for loop closures at each keyframe insertion by matching ORB descriptors against all previous keyframes using brute-force matching. A candidate must pass several verification stages:

- 1) A proximity gate rejects candidates whose raw trajectory distance exceeds 500 m.
- 2) The brute-force match count must exceed 80 after a ratio test.
- 3) PnP-RANSAC verification must produce at least 50 inliers with the same 3.0-pixel threshold.
- 4) The estimated relative translation must be below 100 m to reject implausible matches.

E. Pose-Graph Optimization

When a loop closure is verified, pose-graph optimization is triggered. The optimizer uses SE(3) Lie algebra parameterization (6 parameters per pose: 3 translation + 3 axis-angle rotation) and is solved with a least squares solver. The optimization redistributes the accumulated drift across the full trajectory.

III. EXPERIMENTS AND RESULTS

The system was evaluated using the KITTI Odometry Benchmark (sequences 00–10), which measures translational drift as a percentage and rotational drift in degrees per meter. With an average translational error of 7.61% and a processing speed of 214.7 ms per frame.

The system achieved an average translational drift of 7.61% and a rotational error of 0.031 deg/m across the 23,054 frames of the KITTI benchmark, confirming that the integration of ORB-based tracking and Pose-Graph Optimization provides a reliable framework for autonomous navigation. The highest precision was observed in urban settings, such as sequences 04 and 07, where errors remained below 2%. In longer trajectories, the backend successfully managed cumulative drift through loop closure detection, which “snapped” the map into global alignment upon revisiting known locations.

IV. ANALYSIS

The system performed well in urban sequences, where the combination of ORB feature tracking and stereo-triangulated metric depth provided stable odometry. The low rotational error across most sequences confirms that the PnP-RANSAC motion estimation is resilient against camera tilt and rapid

turns. In sequences where locations are revisited, Pose-Graph Optimization managed to correct accumulated drift. Though drift still accumulates the further we are from the starting location.

Conversely, specific environmental factors introduced significant outliers. Sequence 01 highlighted the limitations of stereo cameras at high speeds on highways, where distant landmarks reduce triangulation accuracy. Also sequence 08 has high translation error since it is never able to detect a loop closure because the vehicle is moving in the opposite direction than the first time.

V. CONCLUSION

The evaluation demonstrates that the Stereo Visual SLAM pipeline is effective for urban navigation. The integration of stereo-based scale recovery and pose-graph optimization allows the system to maintain a high degree of spatial consistency, though performance remains sensitive to environment-specific feature density and vehicle speed. Also we observed that positional drift still accumulates the further we get from the starting point.

VI. APPENDIX

During the project, we ran into a few main technical issues that we had to work around:

- **Slow Execution Time:** Initially, our code was extremely slow, and a full evaluation run took over 3 hours to complete. We spent time profiling and optimizing our tracking logic, and eventually managed to get the run time down to about 6 minutes.
- **Moving Objects:** The code does not currently recognize if objects in the frame are moving. This means that sometimes moving objects, like scooters, are included as static parts of the 3D map. This caused loop closures to fail even when visiting a previously visited location. Fixing it was out of scope for this project.
- **Reverse Loop Closures:** Our stereo visual mapping could not recognize if we were in the same space but driving the opposite way. The loop closure basically only works if we revisit a spot from the same direction, which is limitation of the current system.

VII. REFERENCES

- "PL-SLAM: a Stereo SLAM System through the Combination of Points and Line Segments"
<https://arxiv.org/pdf/1705.09479>
- "LDSO: Direct Sparse Odometry with Loop Closure"
<https://arxiv.org/pdf/1808.01111>"